4. Many-to-Many Relationship

A many-to-many relationship occurs when multiple records in a table are associated with multiple records in another table.

Example 1: Customers and Product

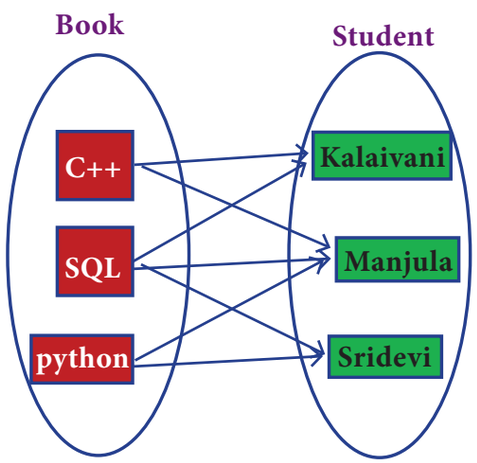
Customers can purchase various products and Products can be purchased by many customers

Example 2: Students and Courses

A student can register for many Courses and a Course may include many students

Example 3: Books and Student.

Many Books in a Library are issued to many students.



Many to Many Relationship
Fig 11.11





The relational model was invented by Edgar Frank Codd (Father of Relational DataBase) as a general model of data, and subsequently promoted by Chris Date and Hugh Darwen among others.

11.9 Relational Algebra in DBMS

What is Relational Algebra?

Relational Algebra, was first created by ***Edgar F Codd*** while at IBM. It was used for modeling the data stored in relational databases and defining queries on it.

Relational Algebra is a procedural query language used to query the database tables using SQL.

Relational algebra operations are performed recursively on a relation (table) to yield an output. The output of these operations is a new relation, which might be formed by one or more input relations.

Relational Algebra is divided into various groups

Unary Relational Operations

SELECT (symbol : σ)

PROJECT (symbol : $\prod$)

Relational Algebra Operations from Set Theory

- UNION ($\cup$)
- INTERSECTION ($\cap$)
- DIFFERENCE ($-$)
- CARTESIAN PRODUCT (X)

SELECT (symbol : σ)

General form σ_c (R) with a relation R and a condition C on the attributes of R.

The SELECT operation is used for selecting a subset with tuples according to a given condition.

Select filters out all tuples that do not satisfy C.

XII Std - CS EM Chapter-11.indd 198 1/24/2020 10:09:35 AM